

Task 2B: peak-day allocation policy (S1, Peliyagoda) · Team AlphaZero

77 of 85 orders served (318.5 of 409.9 m³) on 25 trips with 19 vehicles (4 reefers: 3 trucks + 1 van; 15 ambient: 13 trucks + 2 vans).
check_allocation.py : PASSED. Files: submissions/submission_task2b.csv , analysis in the final notebook §23.

What limits service

- 1. **Refrigerated capacity is the binding resource.** Chilled demand is **181.6 m³** (26 orders, 7 districts). Only 4 reefers are available (trucks VEH003, 006, 007 and van VEH036), **86.2 m³ per wave**, and each has 2 trips inside a 270-minute Fresh window. Distance eats that window: one Puttalam stop is 173 + 15 = **188 min**; three Kurunegala stops 127 + 2×19 + 3×15 = **210**; two Matara stops 137 + 10 + 2×15 = **177**. A reefer sent far cannot make a useful second trip. Solving for maximum chilled volume shows **no plan can deliver more than 143.8 m³**, so **37.9 m³ (21%) of chilled demand is deferred whatever the policy**.
- 2. **Size and access.** S1-078 (Style, Kurunegala, 40.7 m³) is larger than the biggest truck (38 m³) and orders cannot be split: **unavoidable**. The three van-only chilled Colombo orders (1,096 kg) exceed the reefer van's 1,040 kg, so they use both of its trips.
- 3. **Ambient capacity is ample:** 22 ambient trucks and 2 ambient vans are available, and the plan uses only 13 trucks and 2 vans (17 trips). So reefers carry **chilled orders only**. This costs nothing: every ambient order is served except S1-078.

Priority policy

Applied strictly in this order by an exact mixed-integer program; each level is solved to proven optimality, then locked, so a lower rule never trades away a higher one: **(1)** no outlet is deferred two runs in a row; **(2)** most outlets receive their chilled order; **(3)** most chilled m³; **(4)** most total m³; **(5)** outlets waiting longest first; **(6)** fewest trips. Repairs that never change who is served: order each vehicle's trips for least lateness, move late stops to idle trucks, keep each vehicle within one-sixth of its weekly fuel quota (highest: 15.7%).

Segment	Served	Volume served
Fresh ambient	49 / 49	133.0 m³
Tech	5 / 5	18.1 m³
Style	4 / 5	36.6 / 77.2 m³
Fresh chilled	19 / 26	131.0 / 181.6 m³

All 10 orders deferred yesterday are served, including Puttalam OUT074 (5 days without a delivery). Every Fresh outlet whose chilled order waits still receives its ambient groceries.

Deferrals: unavoidable vs our choice, and what they cost

Orders	District	m³	Why	
S1-078	Kurunegala	40.7	larger than any truck	unavoidable
S1-071, 073, 075	Kurunegala	25.0	210 reefer-min for 3 stores; the same minutes serve more stores nearer	choice
S1-064, 067	Matara	12.0	177 reefer-min for 2 stores	choice
S1-021	Colombo	9.9	reefer trip 98% full; only the 7 m³ van has room	choice
S1-056	Galle	3.8	VEH007 at 252 / 270 min; one more stop needs 25	choice

Of the 50.7 m³ of chilled deferrals, 37.9 m³ is unavoidable. The other **12.8 m³ is the price of rule 1**: serving Puttalam (8.7 m³) takes 188 of a reefer's 270 minutes. Maximising chilled m³ alone would deliver 143.8 m³ but leave Puttalam unserved for a 6th day; putting volume before fairness delivers 132.8 m³ and starves the three van-only outlets.

Checks and comparison

- **Beyond the checker:** a real-clock simulation (free-flow travel, handling allowances, waiting for windows, return legs between trips) puts **73 of 77 stops inside their window**. The 4 late stops are chilled stops on second reefer trips (14, 16, 39 and 87 min); a plan that counts return legs still has 3 late stops and delivers 6.7 m³ less, so we keep the published standard, load those trips first and send the stores an ETA. Mall windows are respected.
- **Against simpler rules** (all pass the checker): a greedy priority-score heuristic serves 67 orders (241 m³, 9 of 26 chilled orders) because it fills reefers with ambient orders. Reserving reefers for chilled orders lifts it to 76 orders; exact optimisation adds one more order and 19.7 m³, all chilled.

Recommendations

- **Tomorrow:** the 7 deferred chilled orders become rule-1 priorities; run Kurunegala and Matara first.
- **Bring workshop reefers back before the festival.** VEH004 alone (33.4 m³) serves 23 of 26 chilled outlets (167.9 m³); VEH004 and VEH005 serve all 26 (181.6 m³). Our Task 2A forecast puts New Year week (W15) chilled demand at Peliyagoda near 530 m³, so this is the priority repair.